



Professional Diploma in

Project Management

Defining the Project Plan



Examine a project plan <

Introduce work breakdown structures (WBS) <

Explore project dependencies <

Examine estimation techniques <

Look at diagrams and resource planning <

Objectives

Lesson 3



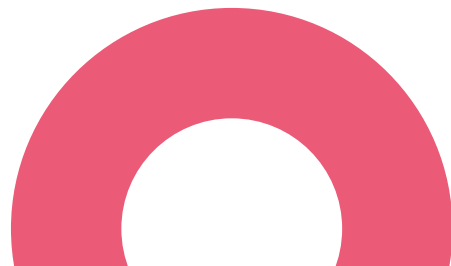
**What is a
project plan?**





Project plan

- Document used to control the execution of the project
- Clearly defines the goals of the project
- Outlines the plan
- Living document – may evolve as the project progresses





»» Project plans include:

- Task list
- Work breakdown structures
- Dependencies outline
- Objectives and constraints
- Resource planning
- Project schedule



Work breakdown structures

Define tasks

Identify & document all activities that need to be performed

Create summary task, subtask, & work packages

Work packages are broken down into activities

Activities are small segments of work





Work breakdown structures

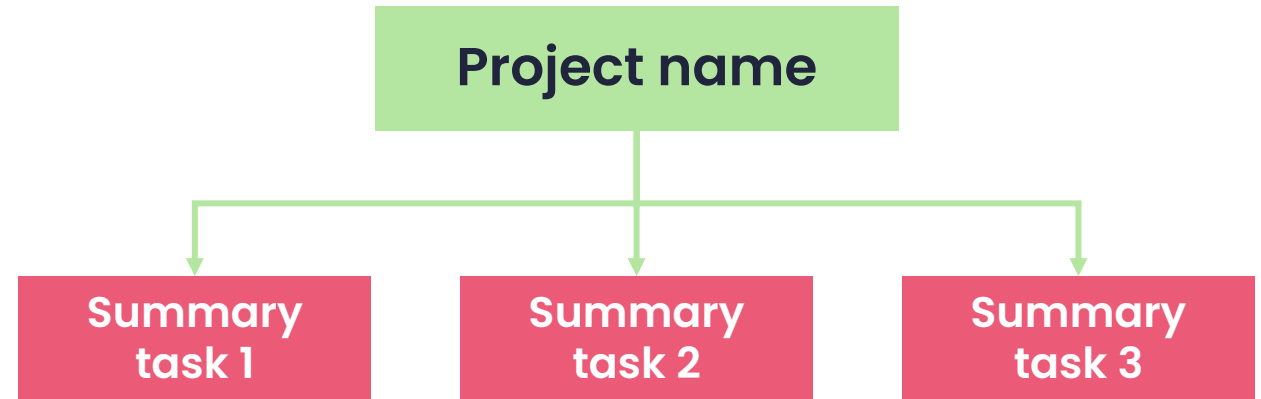
- Break projects into manageable sections
- Sections organised in a hierarchical structure
- Biggest or most important tasks at the top
- More detail as you move down
- Deliverables broken down into sub-deliverables & then work packages





Summary tasks

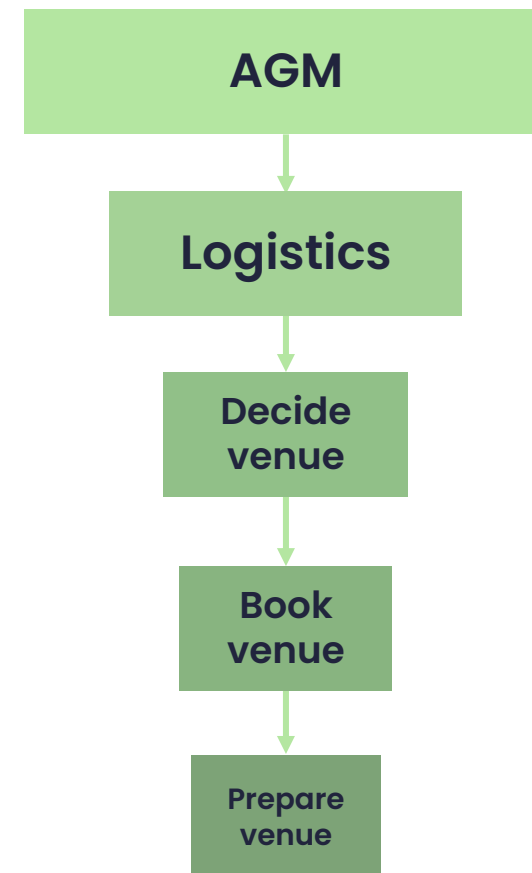
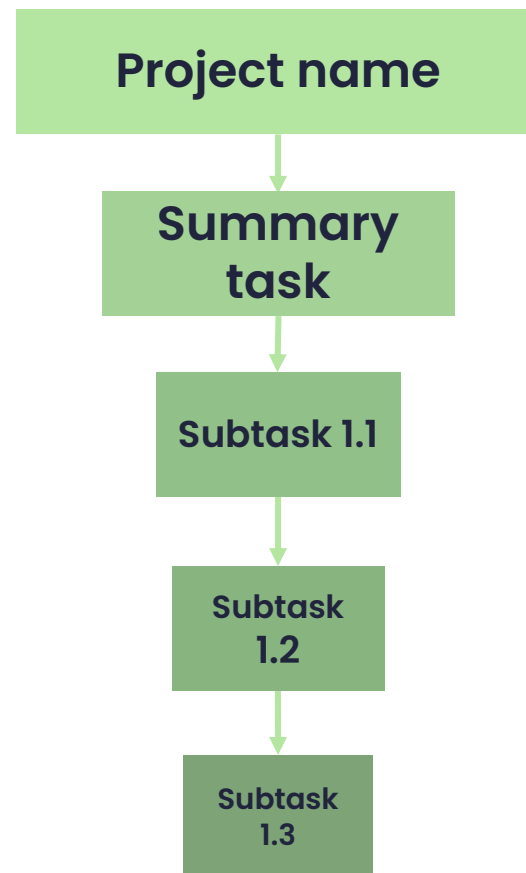
- High-level tasks
- Assist in navigating task groups
- Present milestones, project phases in an easy-to-read format





Sub tasks

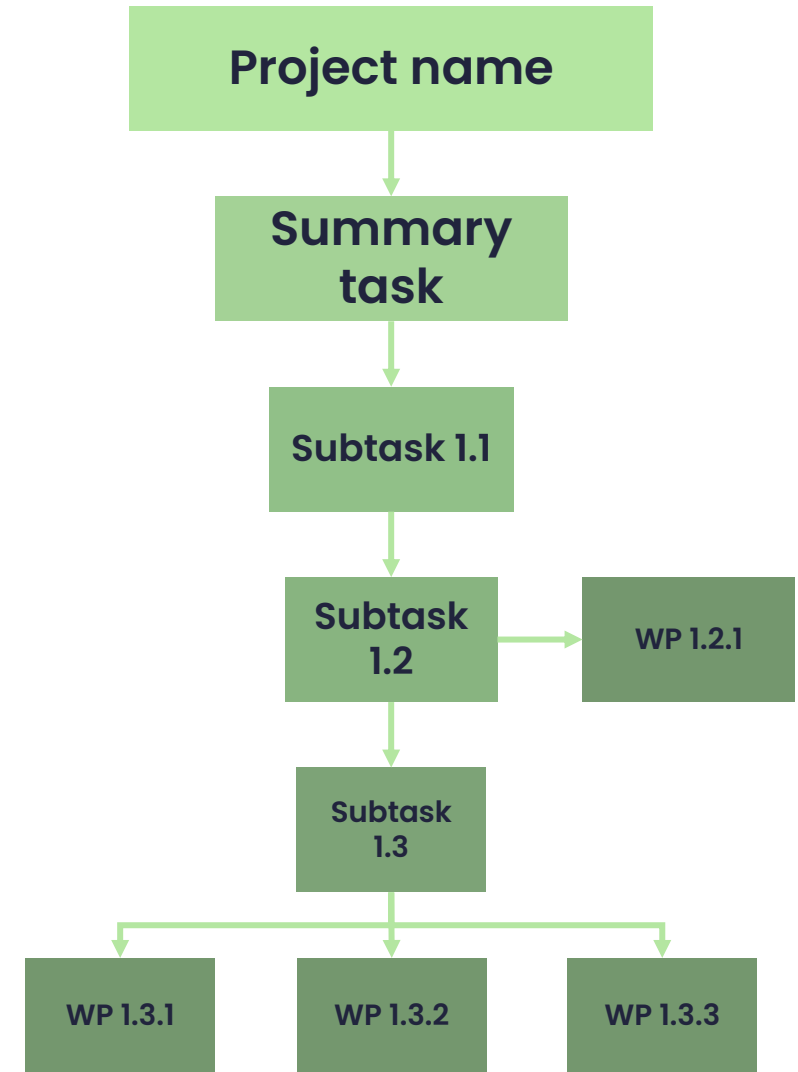
- Break down of summary tasks
- Different activities





Work packages

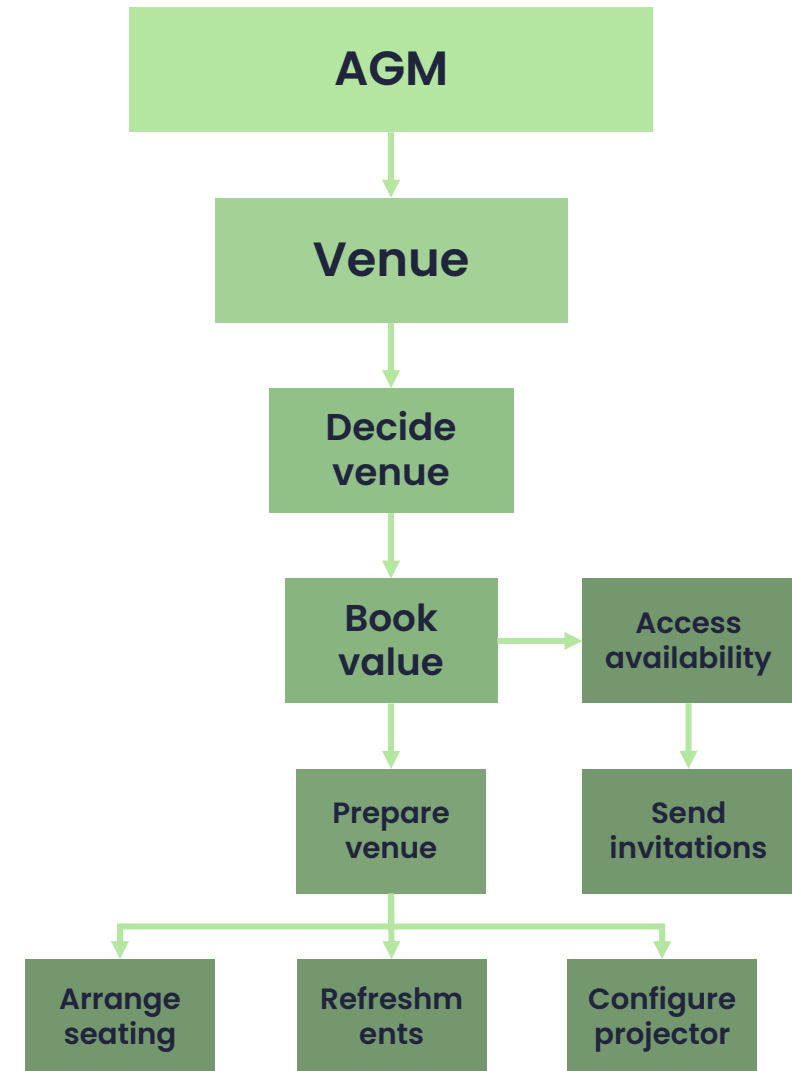
- Allow for large pieces of work to be broken down into more manageable chunks
- Enable multiple teams to focus on different aspects of the project
- Cannot be broken down further
- Often grouped into areas of expertise (technology, human resources, finance)

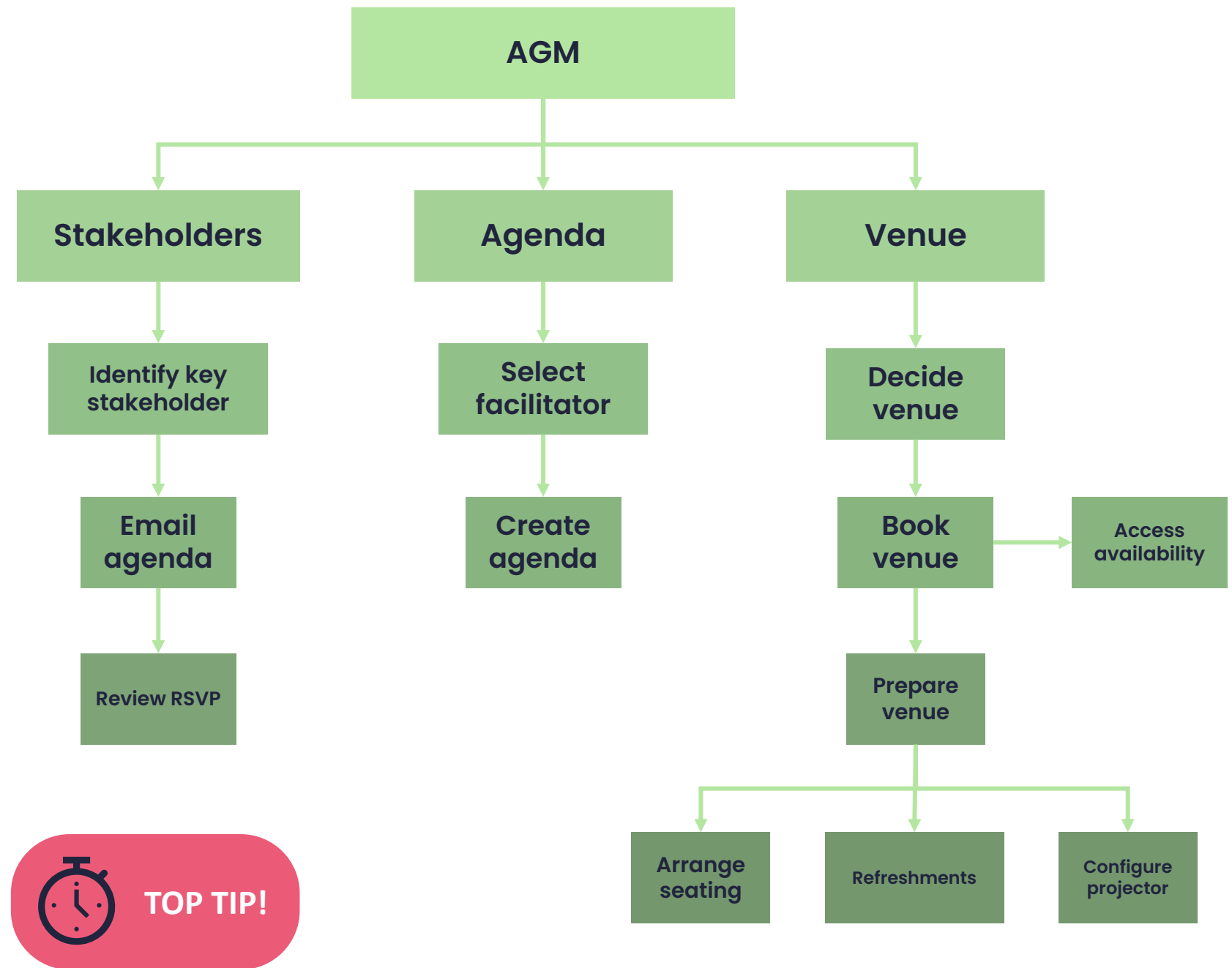




Work package rules

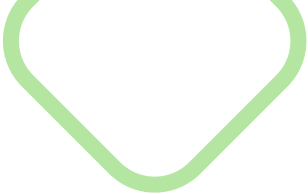
- Should not exceed 10 days
- Each work package should be independent
- Unique & not duplicated







Q & A



When reviewing a WBS which element can't be broken down further?

A

Summary tasks

B

Work packages

C

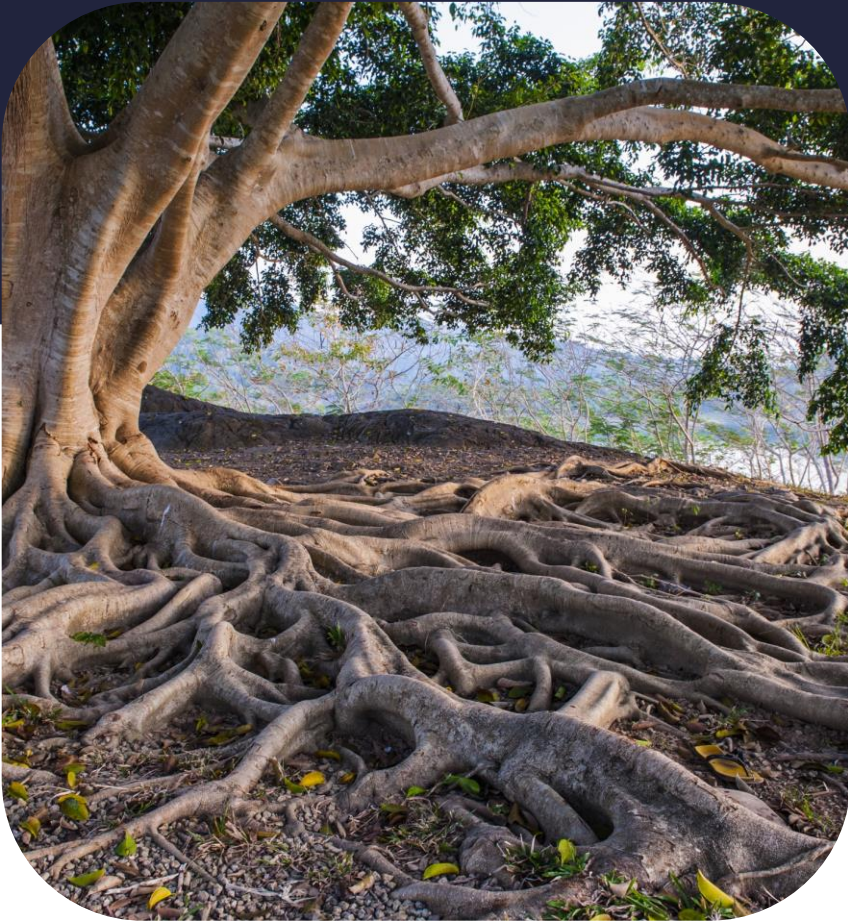
Subtasks





Project dependencies

Outline dependencies



Dependencies are the relationships of the preceding task to the succeeding task

Tasks may have multiple preceding tasks and multiple succeeding tasks

Sequencing of activities

Activity relationship types:

- Finish to Start (FS)
- Start to Start (SS)
- Start to Finish (SF)
- Finish to Finish (FF)

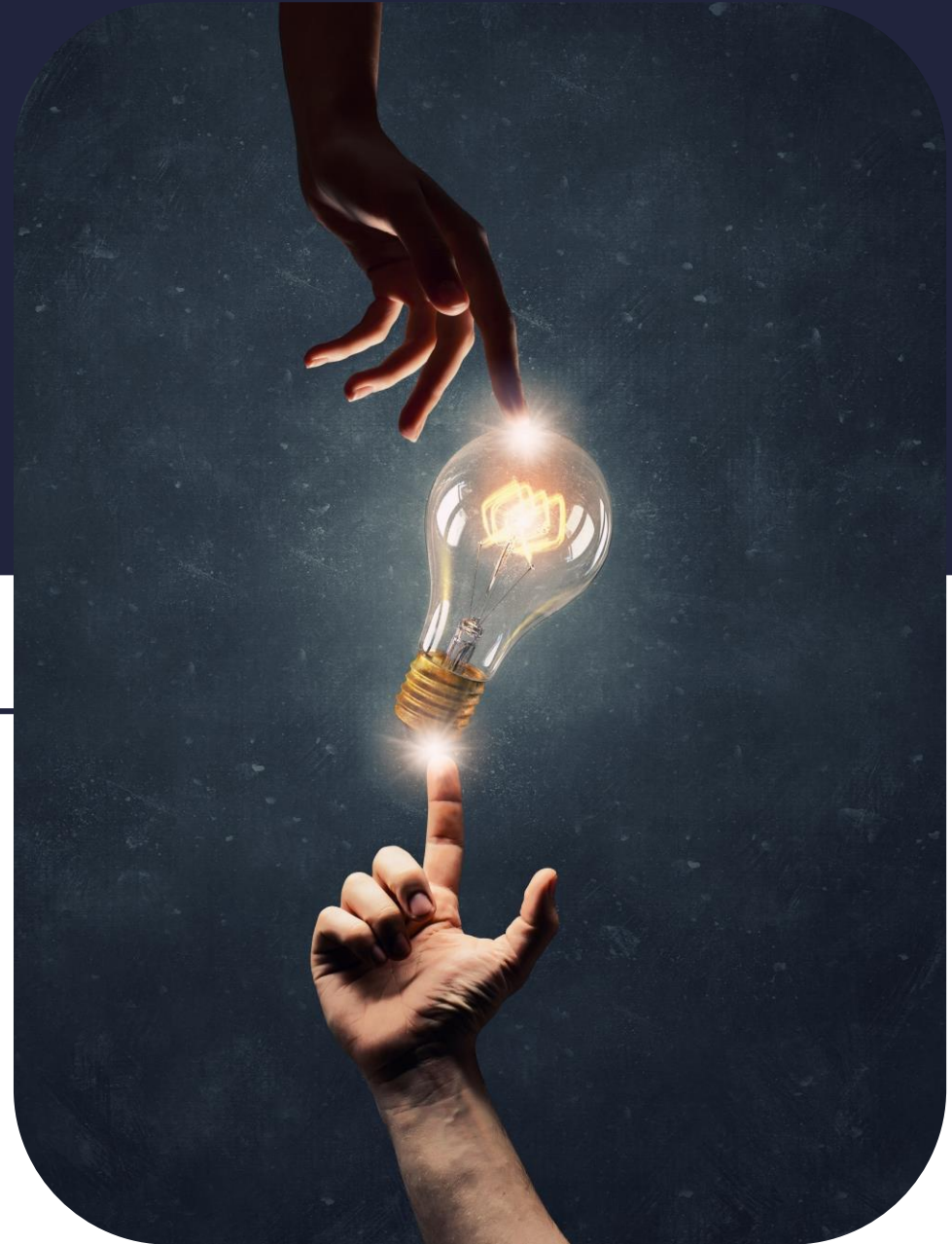


Predecessor activity

Successor activity

Finish

Start

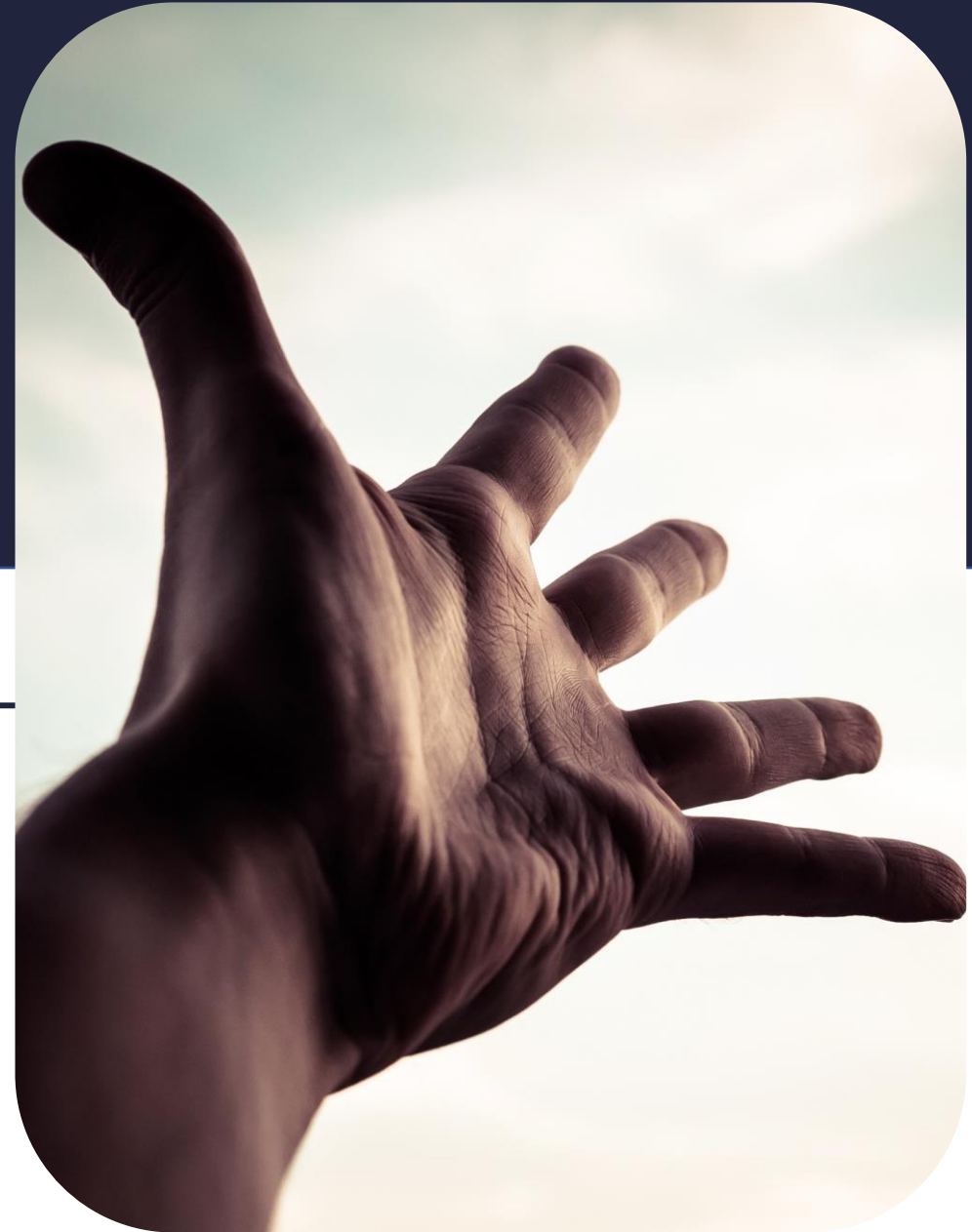


Predecessor activity

Successor
activity

Start

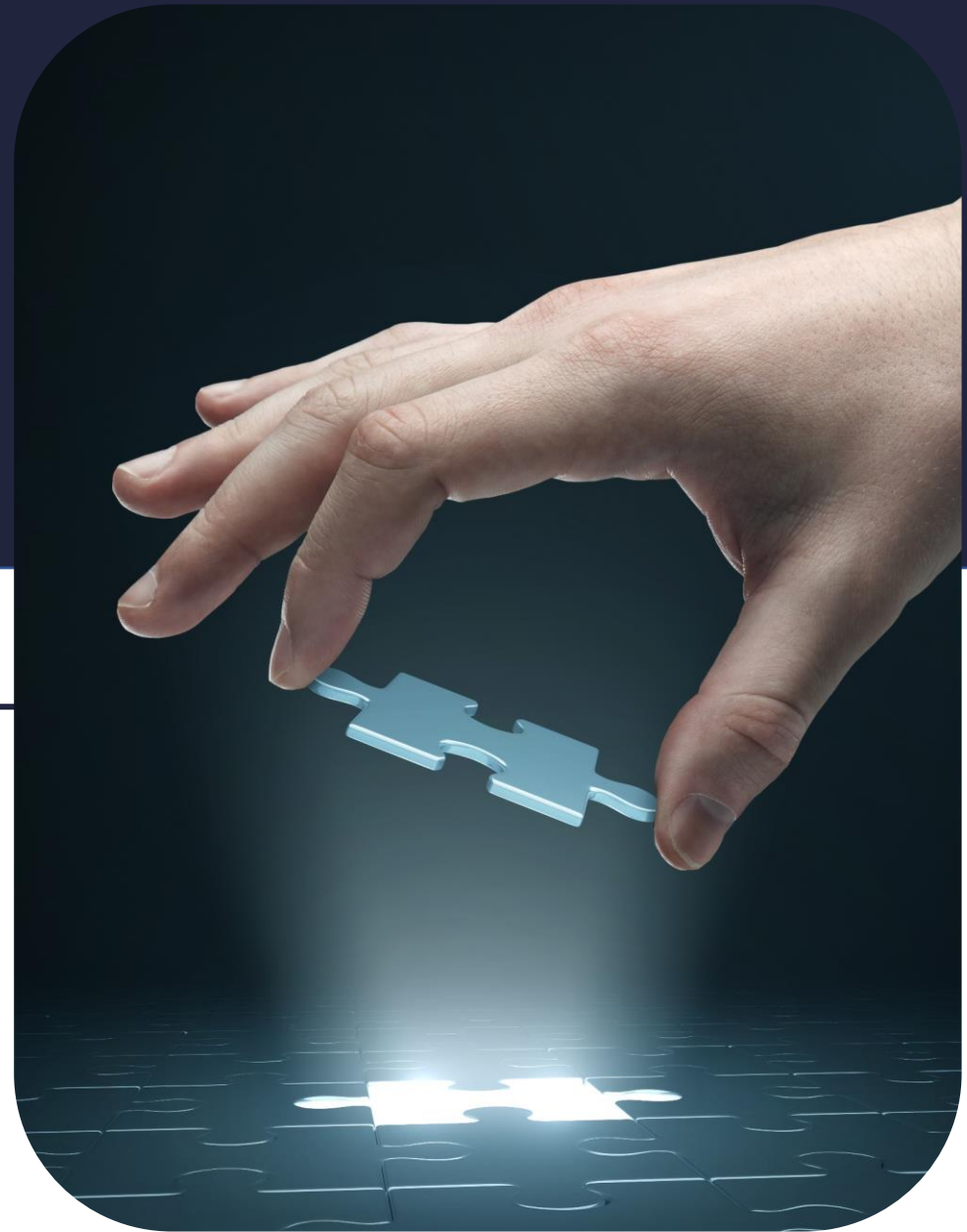
Start





Start

Finish





Finish

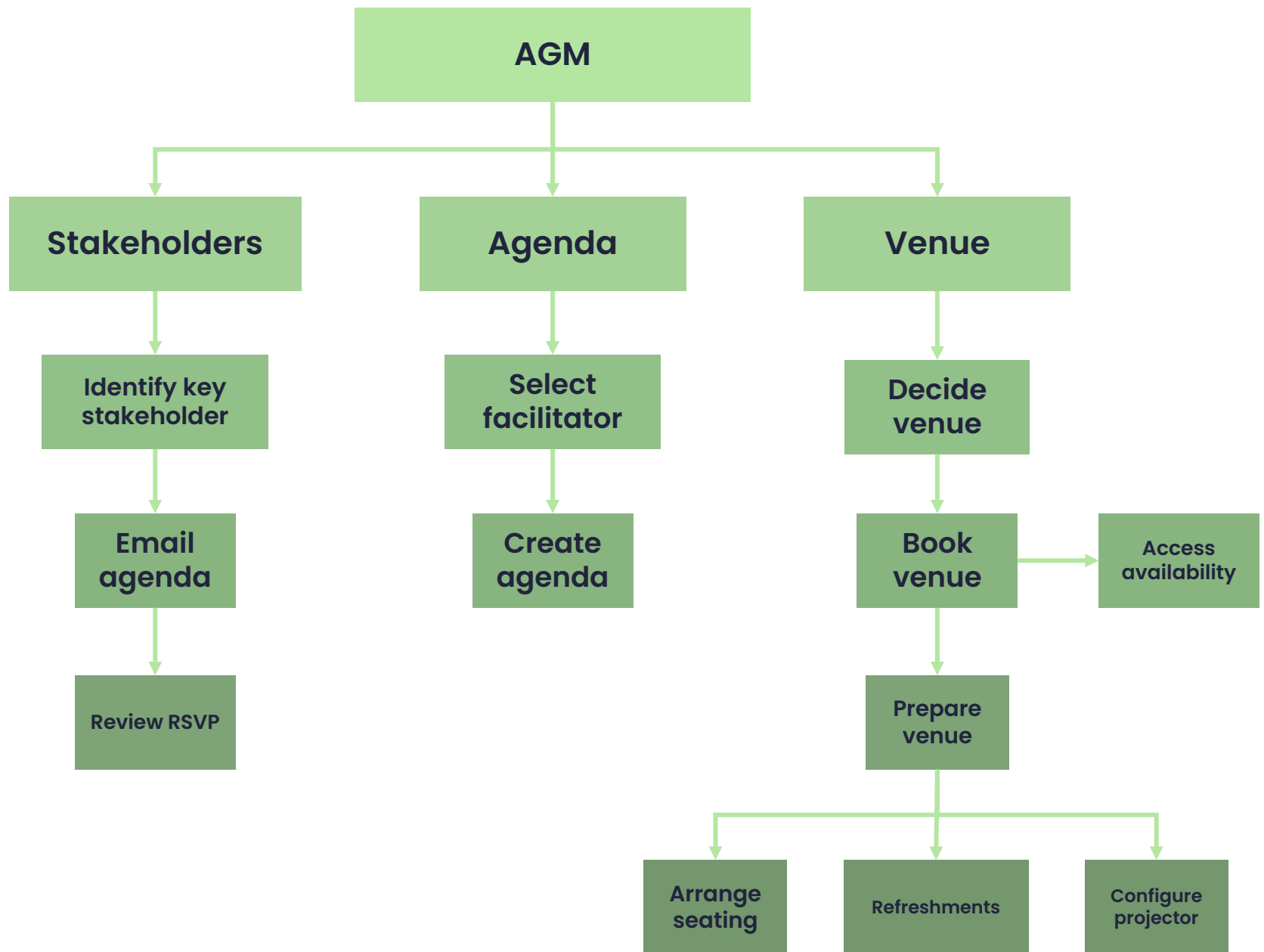
Finish

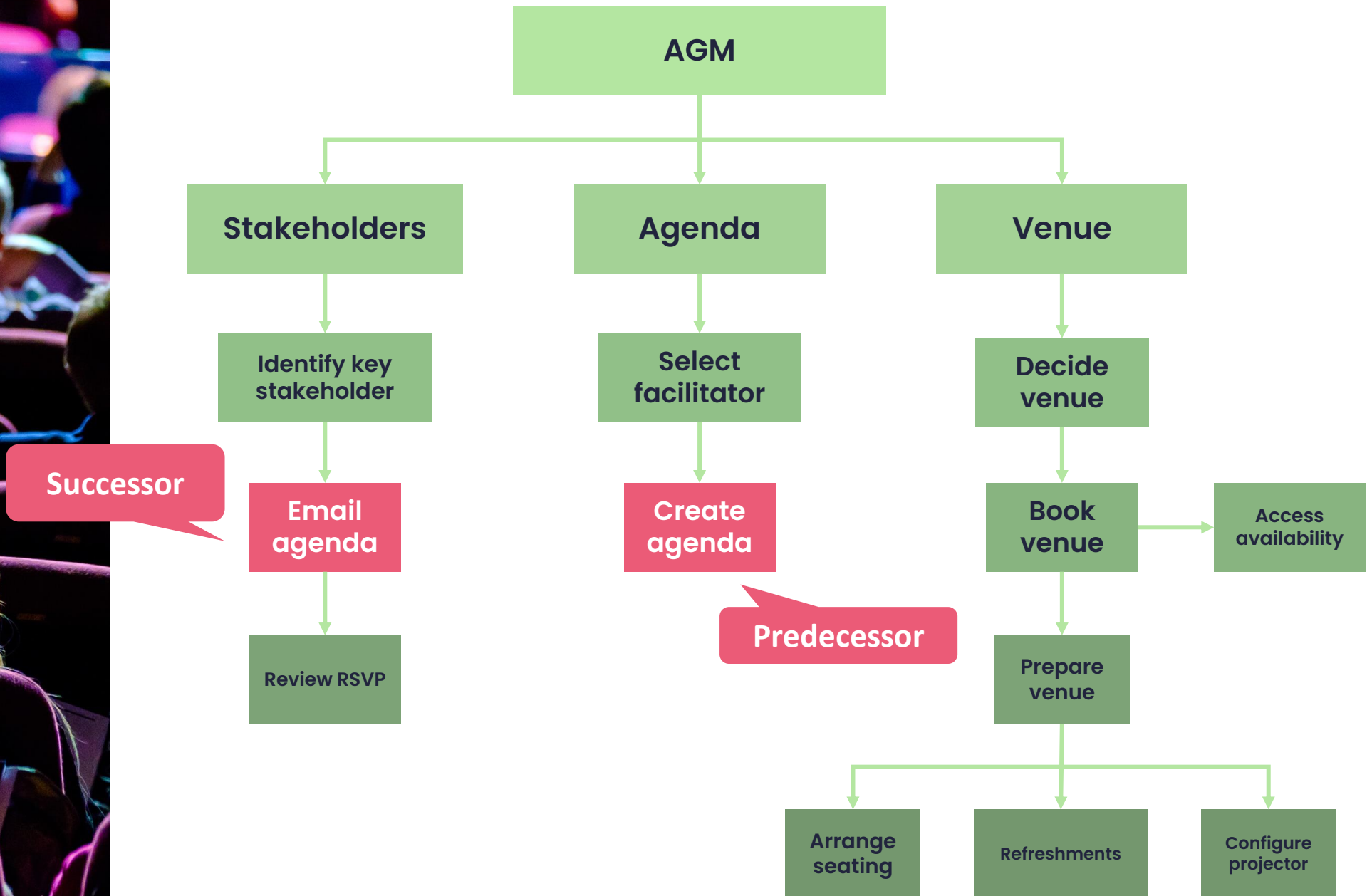


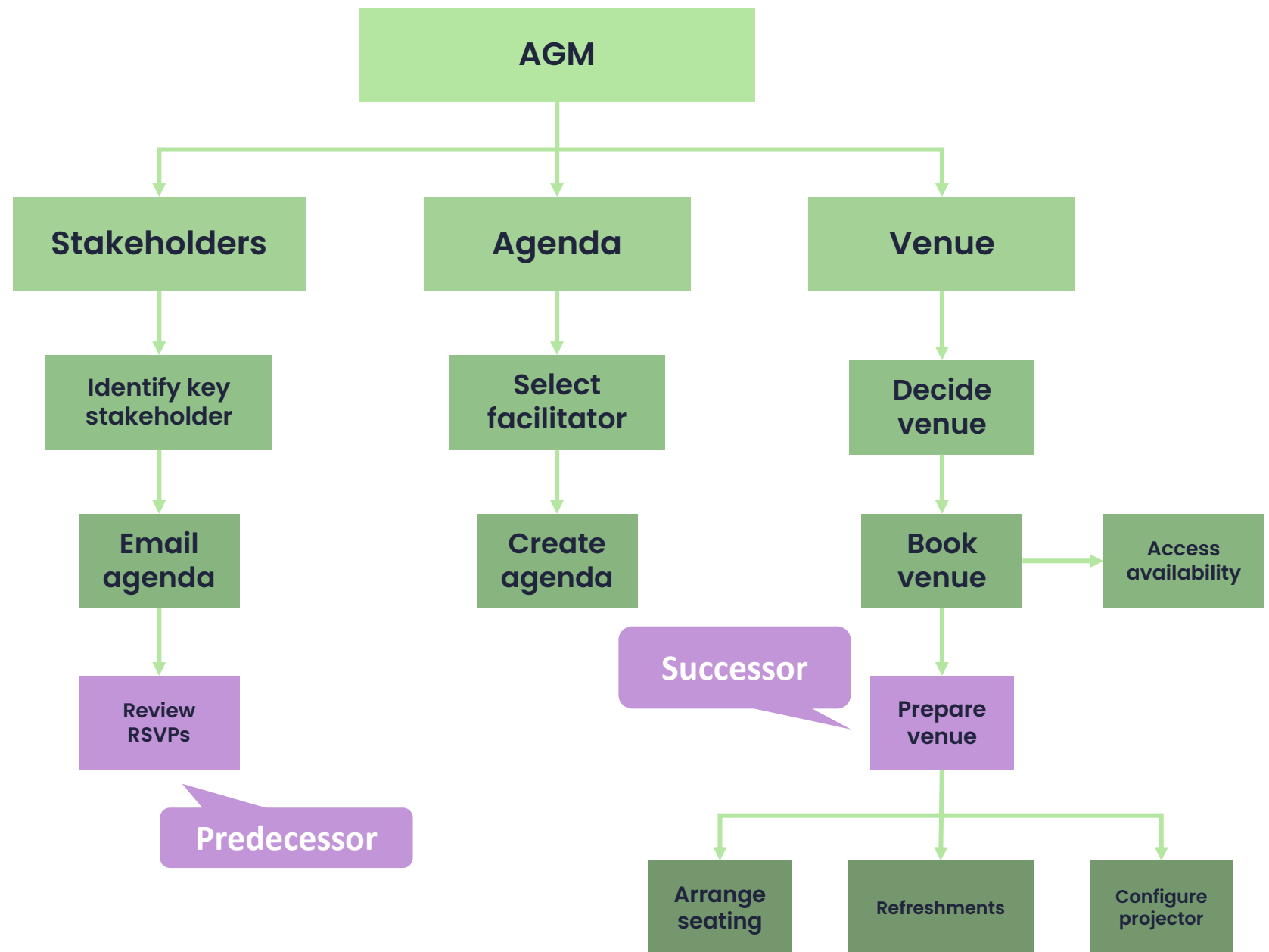


Let's look at the WBS again – did you spot any dependencies?









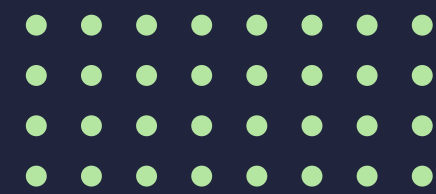


Estimation techniques



»» Estimating activity durations

- Bell curve
- Delphi technique
- Bottom-up



Bell Curve

(Normal distribution)

A bell curve is a graph that is considered to be a normal distribution.

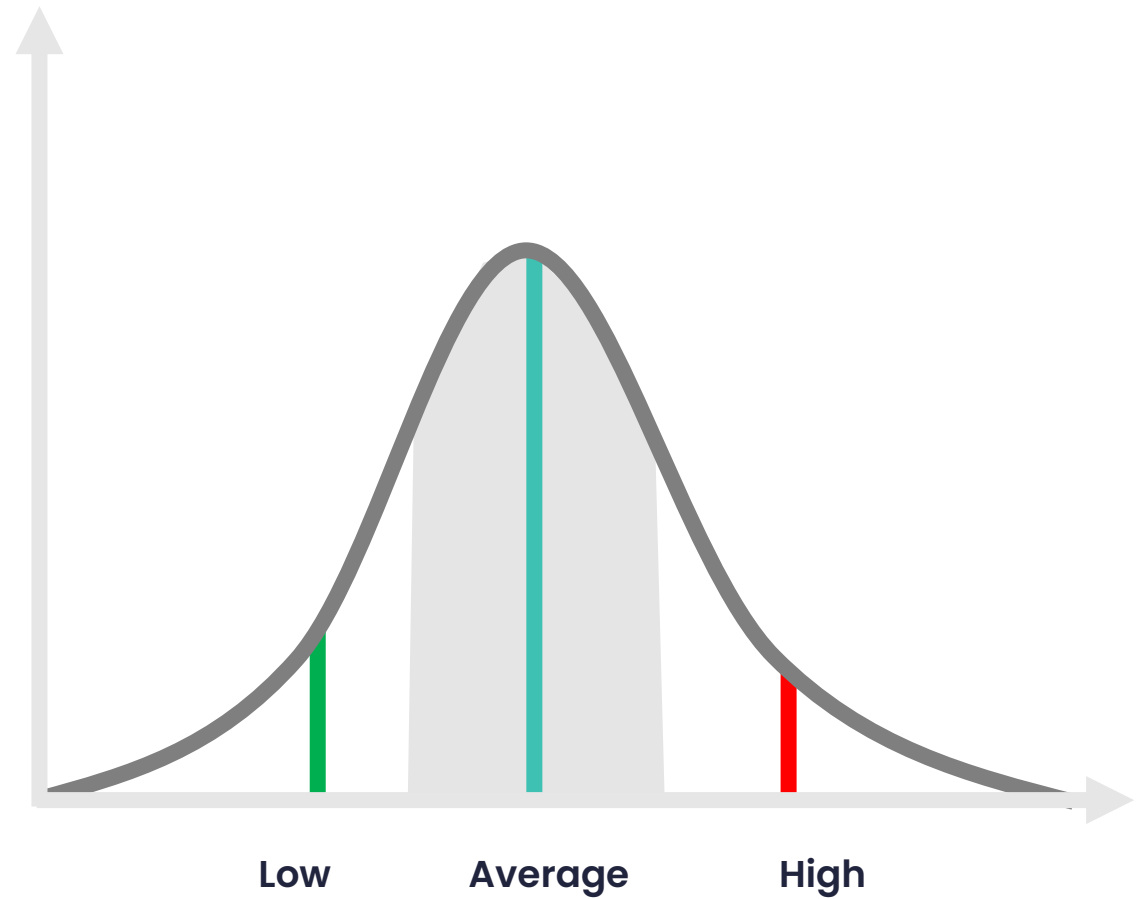
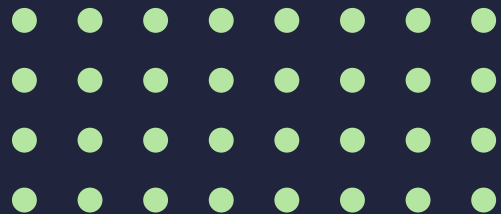
The top of the curve shows the most likely event out of the data collected.

After the mean is calculated, standard deviations are figured.

Standard deviations that depict the returns of a security are known as volatility.

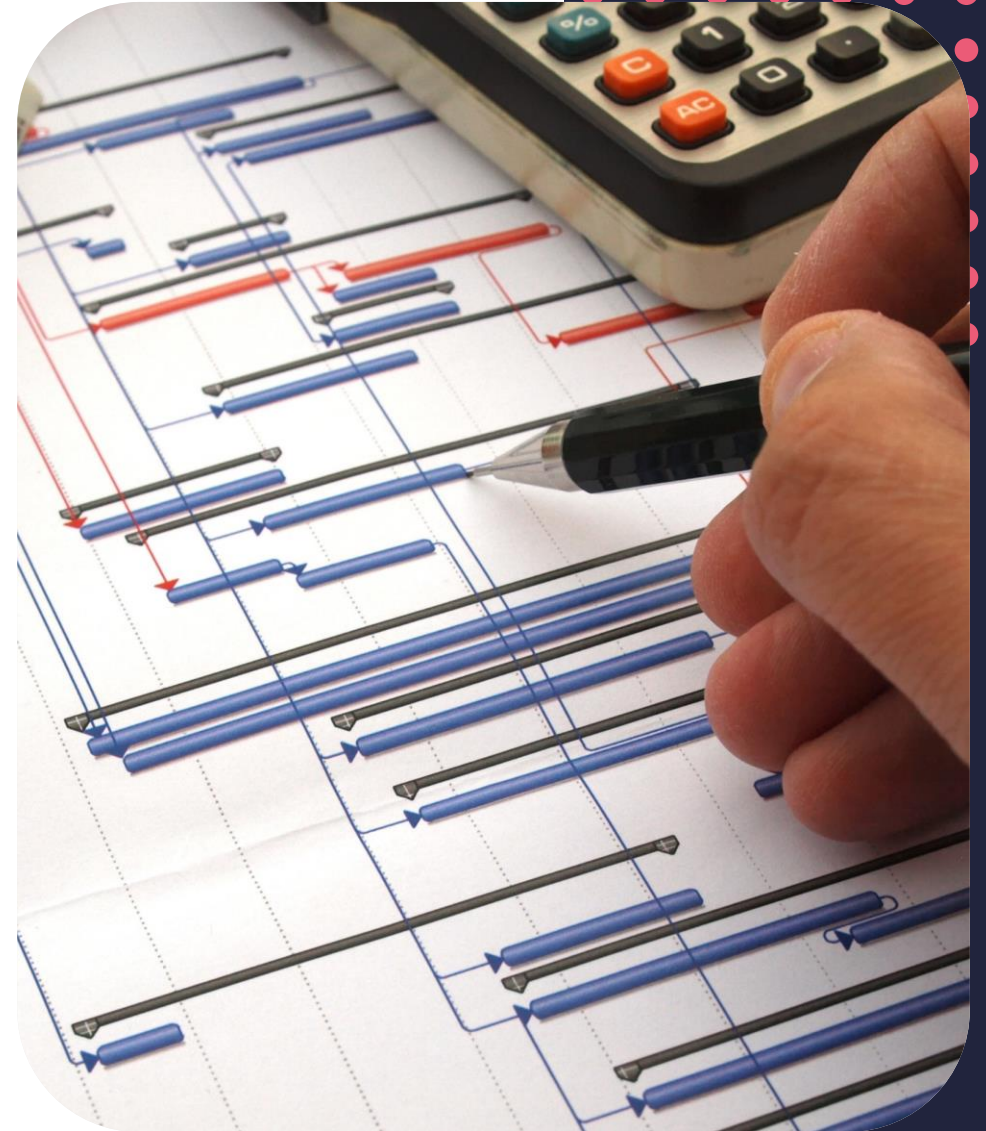


Bell curve technique



Delphi technique

- Experts complete questionnaires individually
- Answers are collated and distributed for all to revise their estimates
- Process can occur many times
- Final estimate based on the average estimate





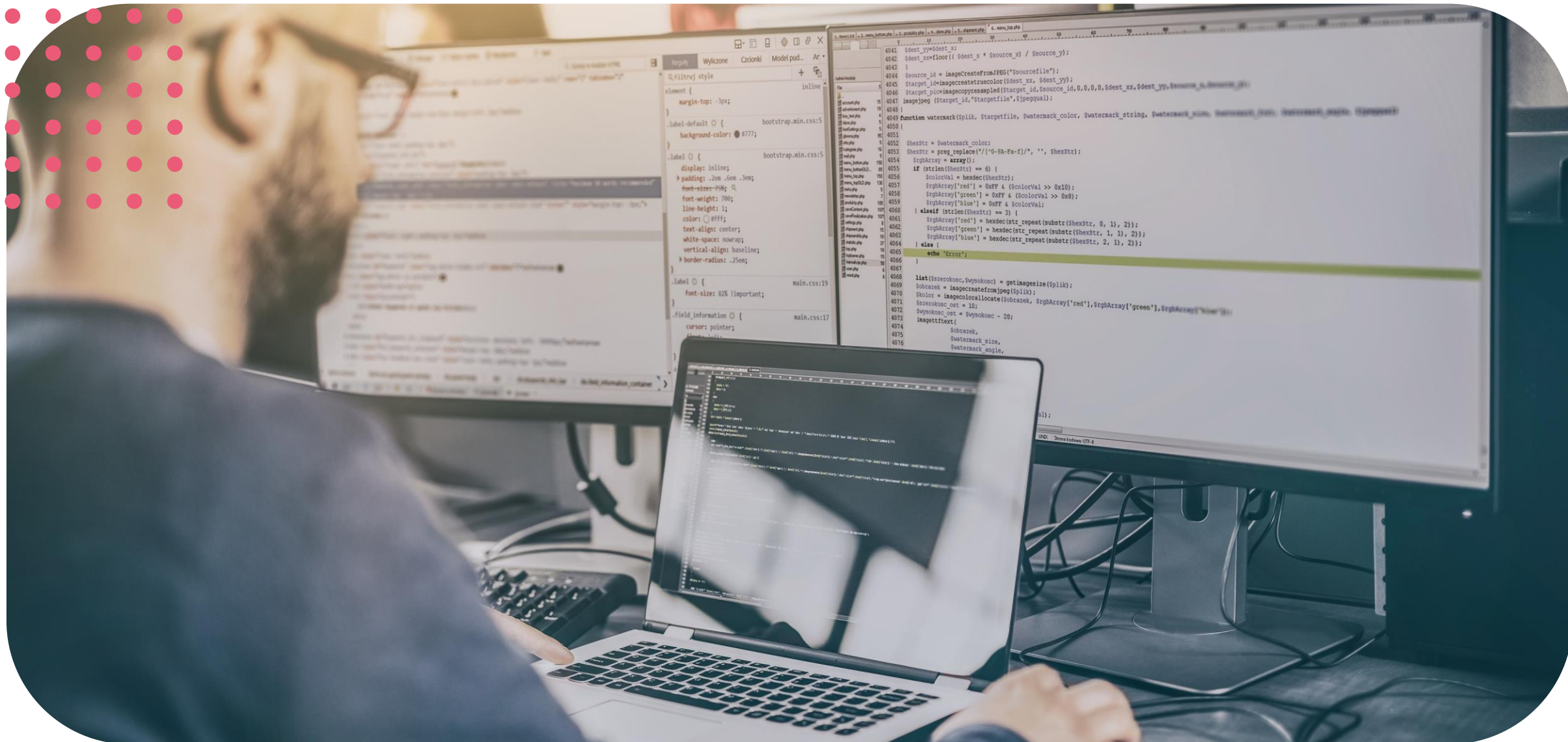
Bottom-up technique

- Project broken down into smaller manageable sections
- Staff provide a time estimate
- Overall estimate calculated as an accumulation of the smaller tasks



Helpful diagrams and resource planning

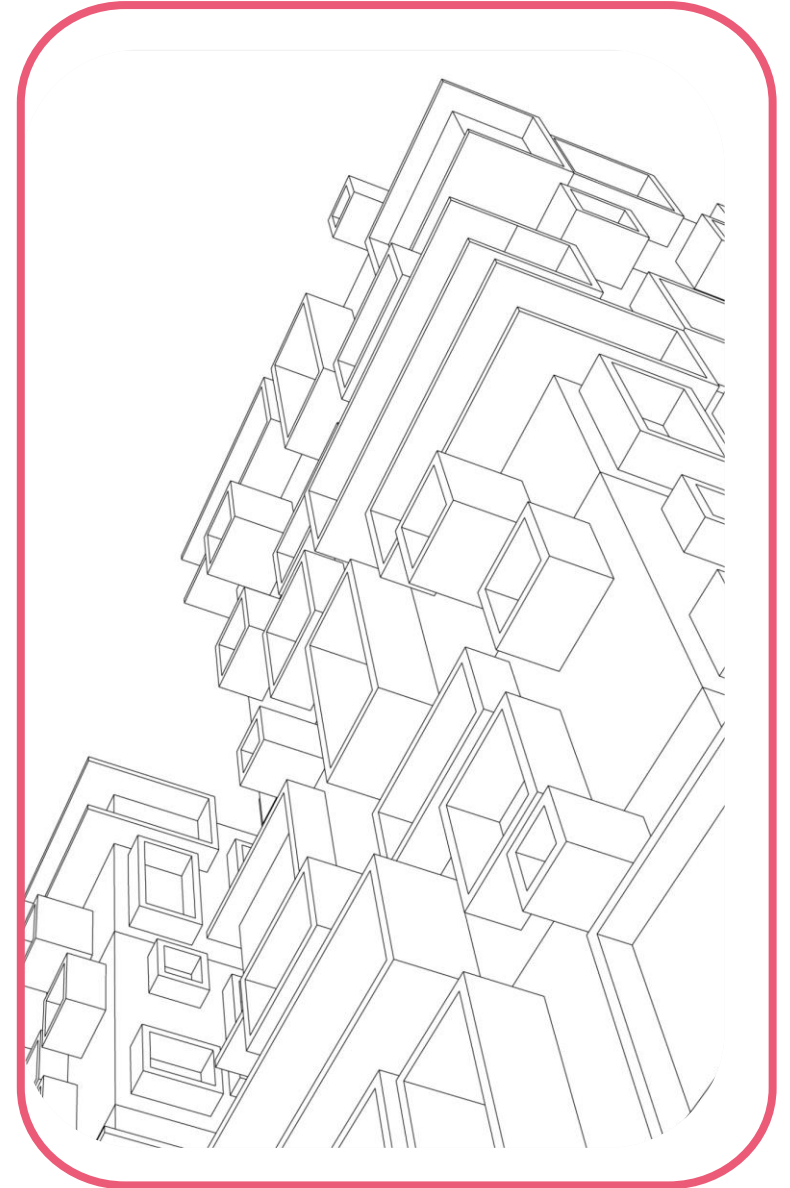
Project calendars





Resource calendar

- Used to plan, manage and monitor employees and equipment
- Details working and non-working times of resources and staff
- Resources will vary depending on the project
- Various resource scheduling tools available



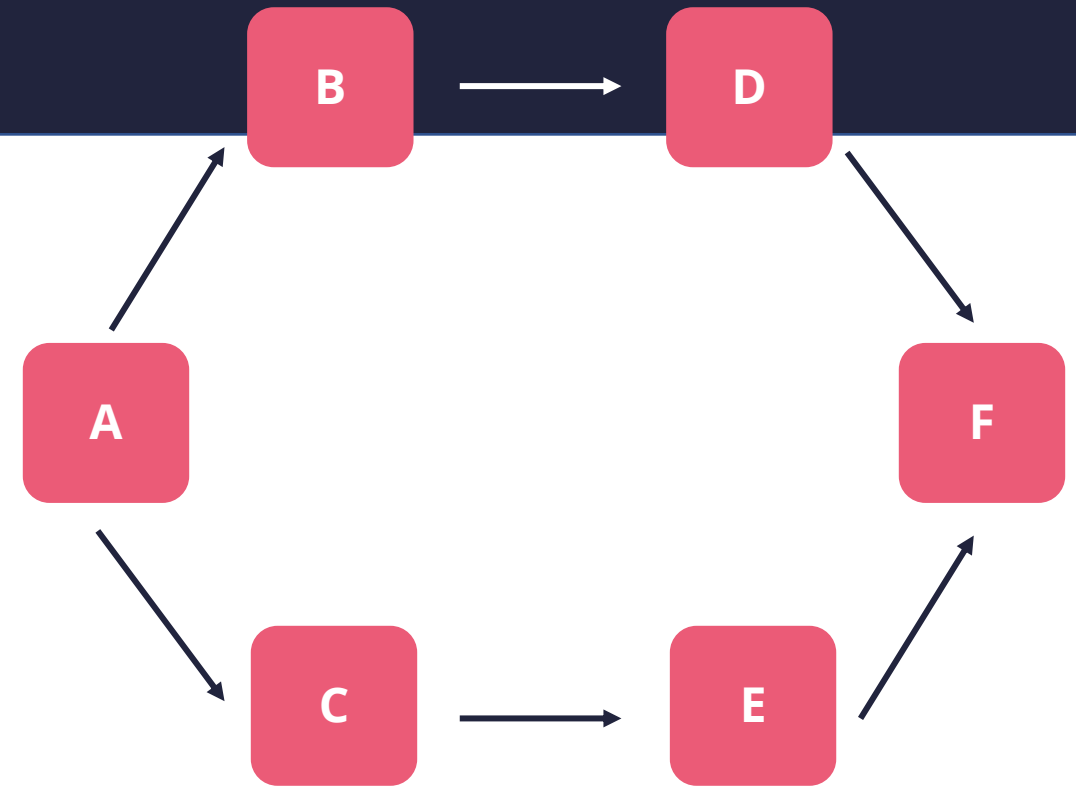


Company calendar

- Company calendars enable the detailing and sharing of segments of time and activities
- Should be easy to use, simple to share, customisable, and available for multiple devices.

Network diagram for setting up a website

Activity	Description	Duration in hours	Preceding Activity
A	Register domain name	2	-
B	Design website	10	A
C	Code website	36	A
D	Web hosting	8	B
E	Copyright	3	C
F	Go live	3	E,D





Task list for setting up a website

Activity	Description	Duration in hours	Team Member(s) assigned
A	Register domain name	2	Sue
B	Design website	10	Claire
C	Code website	36	Thabo
D	Web hosting	8	Rajesh
E	Copyright	3	Malika
F	Go live	3	Carlo



Important fact...

**The absence of
stakeholder buy-in
can easily turn a
multi-million dollar
investment into a
very expensive
lesson.**





Develop a resource plan

Task	Name	Start Date	End Date	Duration
Register a domain name	Sue	02/11	02/11	2



Develop a resource plan

Task	Name	Start Date	End Date	Duration
Register domain name	Sue	02/11	02/11	2
Design website	Claire	02/11	03/11	10



Develop a resource plan

Task	Name	Start Date	End Date	Duration
Register domain name	Sue	02/11	02/11	2
Design website	Claire	02/11	03/11	10
Code website	Thabo	04/11	09/11	36



Develop a resource plan

Task	Name	Start Date	End Date	Duration
Register domain name	Sue	02/11	02/11	2
Design website	Claire	02/11	03/11	10
Code website	Thabo	04/11	09/11	36
Copyright	Rajesh	10/11	10/11	8
Host Website	Malika	11/11	11/11	3
Go Live	Carlo	11/11	11/11	3



Develop a resource plan

Task	Name	Start Date	End Date	Duration
Register domain name	Sue	02/11/2020	02/11/2020	2
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Host website	Malika	11/11/2020	11/11/2020	3

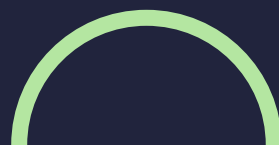


Develop a resource plan

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Copyright	Rajesh	10/11/2020	10/11/2020	8
Host website	Malika	11/11/2020	11/11/2020	3
Go live	Carlo	11/11/2020	11/11/2020	3



Developing a project schedule





Project schedule

- Monitor the status of project activities (RAG)
- Implement changes/solutions for activities that have exceeded expected duration estimate
- Recognise potential risks and plan

Let's talk about software





Suggested software

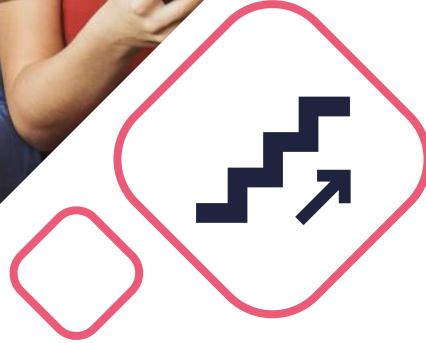
Wrike

Hub Planner

Celoxis

TeamGantt





Lesson Challenge

Research the project planning and scheduling tools available to find the ones that will work best for you.